Consistency Verification of Hypothetical Character Backstories

TRACK A

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1 Abstract

Verifying the consistency of hypothetical character backstories against long narrative documents presents a challenge that conventional Retrieval-Augmented Generation (RAG) systems are not designed to handle. Unlike factual texts, novels encode meaning through distributed events, implicit motivations, temporal dependencies, and thematic development spanning hundreds of pages. Standard RAG pipelines assume localized evidence, explicit contradictions, and lexical relevance, leading to unreliable judgments when applied to narrative fiction.

In this work, we propose an iterative, memory-driven narrative reasoning framework for long-context claim verification. The system decomposes novels into overlapping chunks, retrieves evidence iteratively, and accumulates relevant context in a persistent memory workspace. Reasoning is performed exclusively over this evolving memory rather than isolated passages or the full document. By modeling narrative understanding through veridical, semantic, and episodic memory representations, the framework supports global reasoning across chapters, motivations, and timelines. This approach aligns with benchmarks designed to require non-local, multi-step narrative inference and demonstrates clear advantages over static RAG baselines.

2 Objectives

2.1 Overview of problem

The goal of this project is to verify claims against a long narrative document (novel).
Unlike factual articles, novels contain:

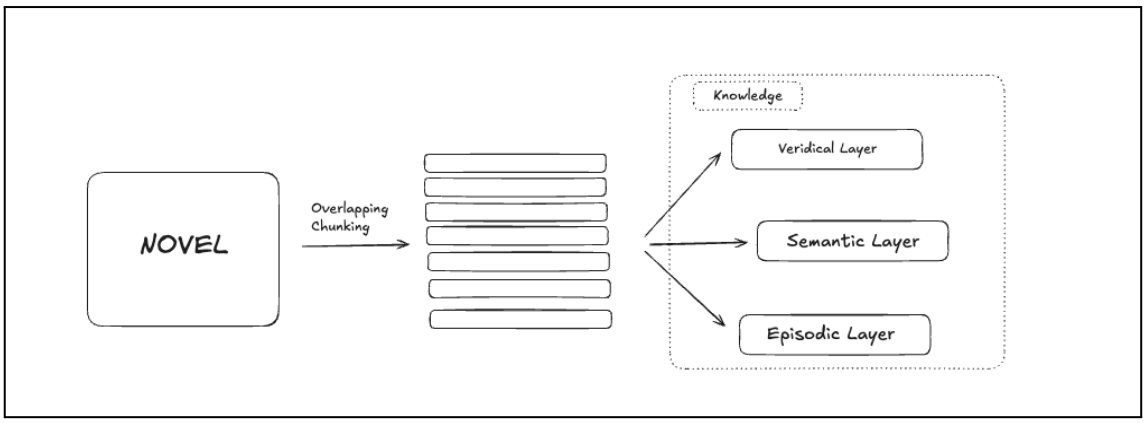
- events spread across chapters
- motives and consequences not stated explicitly
- descriptions, dialogue, filler text

Standard RAG pipelines implicitly assume that

- Relevant evidence is localzsed.
- Semantic similarity is sufficient for correctness. None of these assumptions hold in narrative fiction.

As a result, flat retrieval followed by one-shot reasoning over a small set of retrieved passages is insufficient for global narrative consistency verification. This motivates the need for a fundamentally different reasoning paradigm.

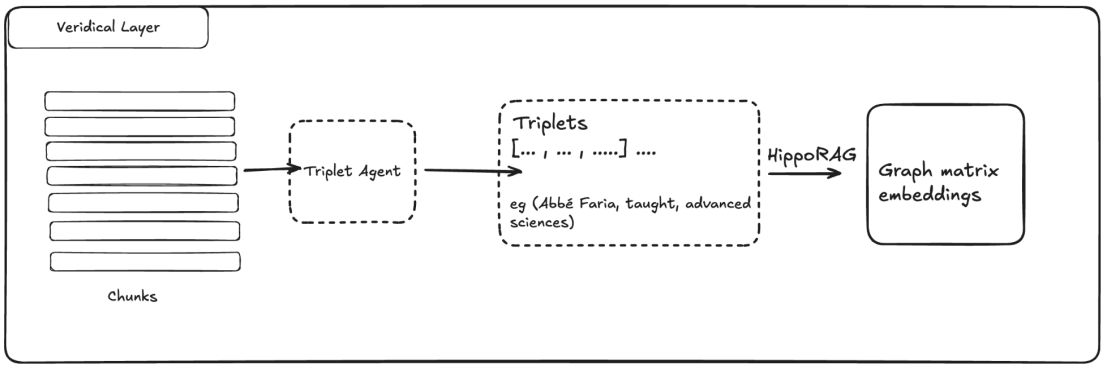
Our approach reframes narrative consistency verification as a **cognitive, iterative, memory-driven reasoning process**, inspired by how humans read and reason about novels. Instead of treating the document as a static source of retrievable facts, we model reasoning as a process that incrementally accumulates, refines, and evaluates evidence across the entire narrative.



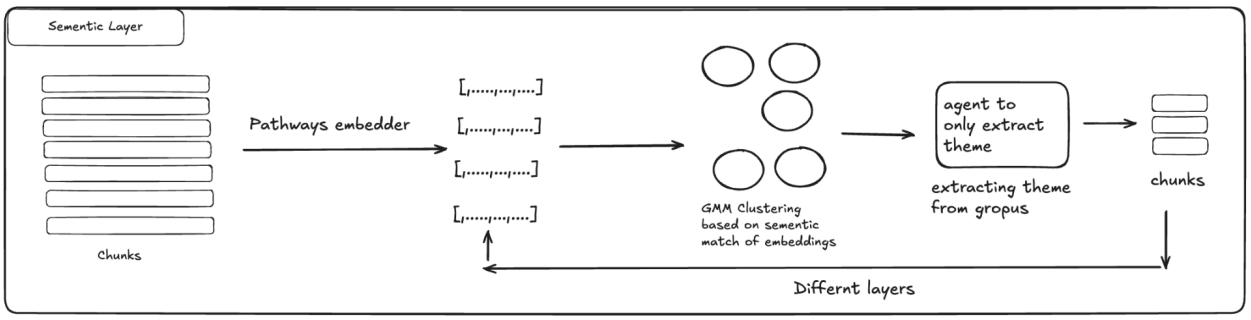
At the core of our system is the observation that human narrative comprehension relies on **multiple complementary memory systems**, each serving a distinct cognitive function. We formalize this intuition into three explicit layers:

- **Veridical Memory** – captures what explicitly happened in the narrative, supporting factual verification and cross-chapter consistency

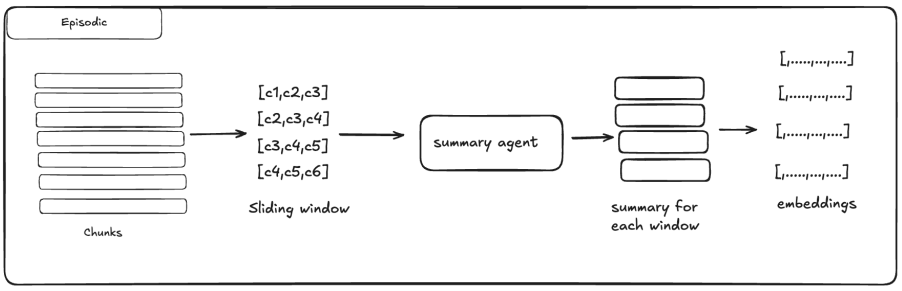
.



- **Semantic Memory** – captures what events and actions mean, including character traits, motivations, and thematic roles.



- **Episodic Memory** – captures when and why events occurred, preserving temporal order and causal dependencies.



Each memory type addresses a specific failure mode of standard RAG: veridical memory resolves distributed factual evidence, semantic memory detects essence-level and motivational mismatches, and episodic memory enforces temporal and causal feasibility.

a. Overall System Architecture

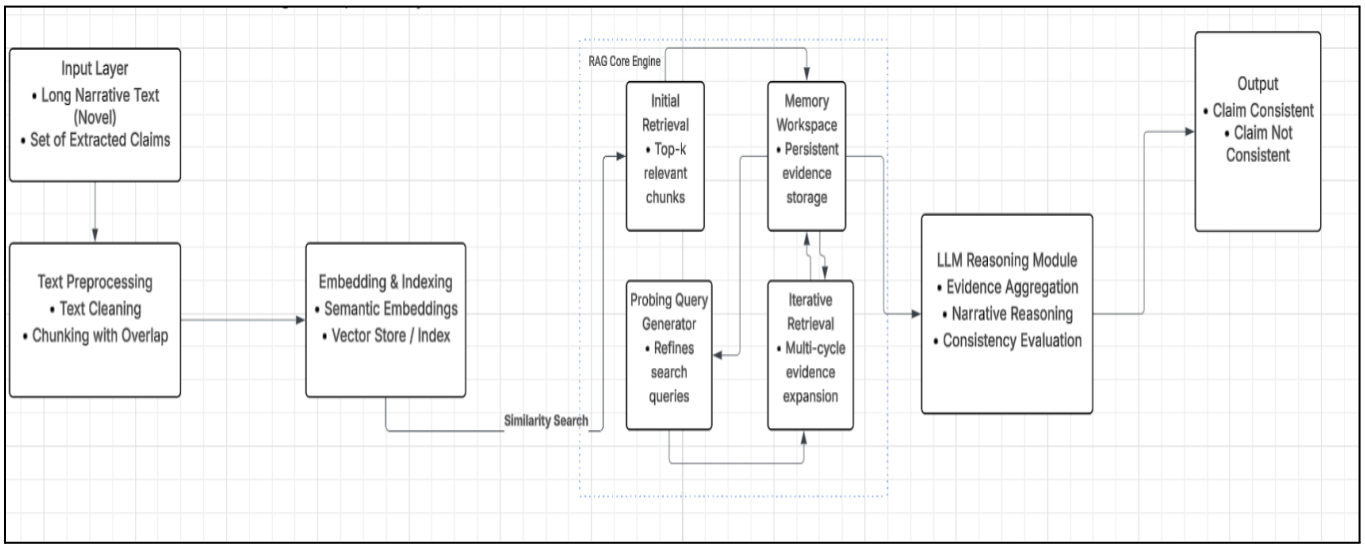


Figure 1: Overall Systemic Pipeline Architecture Diagram

Figure 1 Explanation:

Step 1: Input Layer
Inputs:

- Full novel (long narrative text)
- Set of extracted claims

Step 2: Preprocessing & Chunking

- The novel is divided into overlapping chunks
- Chunking ensures:
- Manageable context sizes

- Partial preservation of narrative continuity

Step 3: Embedding & Vector Index

- Each chunk is converted into a semantic embedding
- All embeddings are stored in a vector index
- This allows similarity-based retrieval for any claim or probe

Step 4: RAG engine
It consists of:

- Initial retrieval based on the claim
- Memory workspace that stores relevant chunks
- Probing query generation when evidence is insufficient
- Iterative retrieval cycles that expand and refine memory

Unlike standard RAG, retrieved evidence is not discarded after one step.

Step 5: LLM-Based Reasoning

- The LLM reasons over accumulated memory, not a single chunk
- It evaluates whether the claim is consistent with the narrative
- Reasoning is contextual, not purely factual

Step 6: Output
Each claim is labelled as:

- Consistent or not Consistent
- Along with its rationale evidence.

b. Core Idea: Multi-Memory, Iterative Narrative Reasoning

Our approach reframes narrative consistency verification as a **cognitive, iterative, memory-driven reasoning process**, inspired by how humans read and reason about novels. Instead of treating the document as a static source of retrievable facts, we model reasoning as a process that incrementally accumulates, refines, and evaluates evidence across the entire narrative.

At the core of our system is the observation that human narrative comprehension relies on **multiple complementary memory systems**, each serving a distinct cognitive function. We formalize this intuition into three explicit memory types:

- **Veridical Memory** – captures what explicitly happened in the narrative, supporting factual verification and cross-chapter consistency.
- **Semantic Memory** – captures what events and actions mean, including character traits, motivations, and thematic roles.
- **Episodic Memory** – captures when and why events occurred, preserving temporal order and causal dependencies.

Semantic consistency is evaluated using probabilistic clustering over abstracted character traits, allowing soft semantic judgments rather than rigid matching thereby clustering with Gaussian Mixture Models (GMM).

Each memory type addresses a specific failure mode of standard RAG: veridical memory resolves distributed factual evidence, semantic memory detects essence-level and motivational mismatches, and episodic memory enforces temporal and causal feasibility.

c. **Iterative Retrieval with Persistent Memory**

Rather than performing retrieval once and discarding evidence, our system retrieves evidence **iteratively** and stores it in a persistent memory workspace. If the accumulated evidence is insufficient to reach a confident decision, the system generates targeted probing queries to retrieve additional context, progressively refining its understanding of the narrative.

This iterative loop mirrors human reading behavior: readers re-examine earlier passages, seek clarifying context, and update their mental model of the story before reaching a judgment. The Large Language Model (LLM) operates exclusively over the accumulated memory rather than the full novel, enabling global reasoning despite limited context windows.

2.2 Handling of Long Context

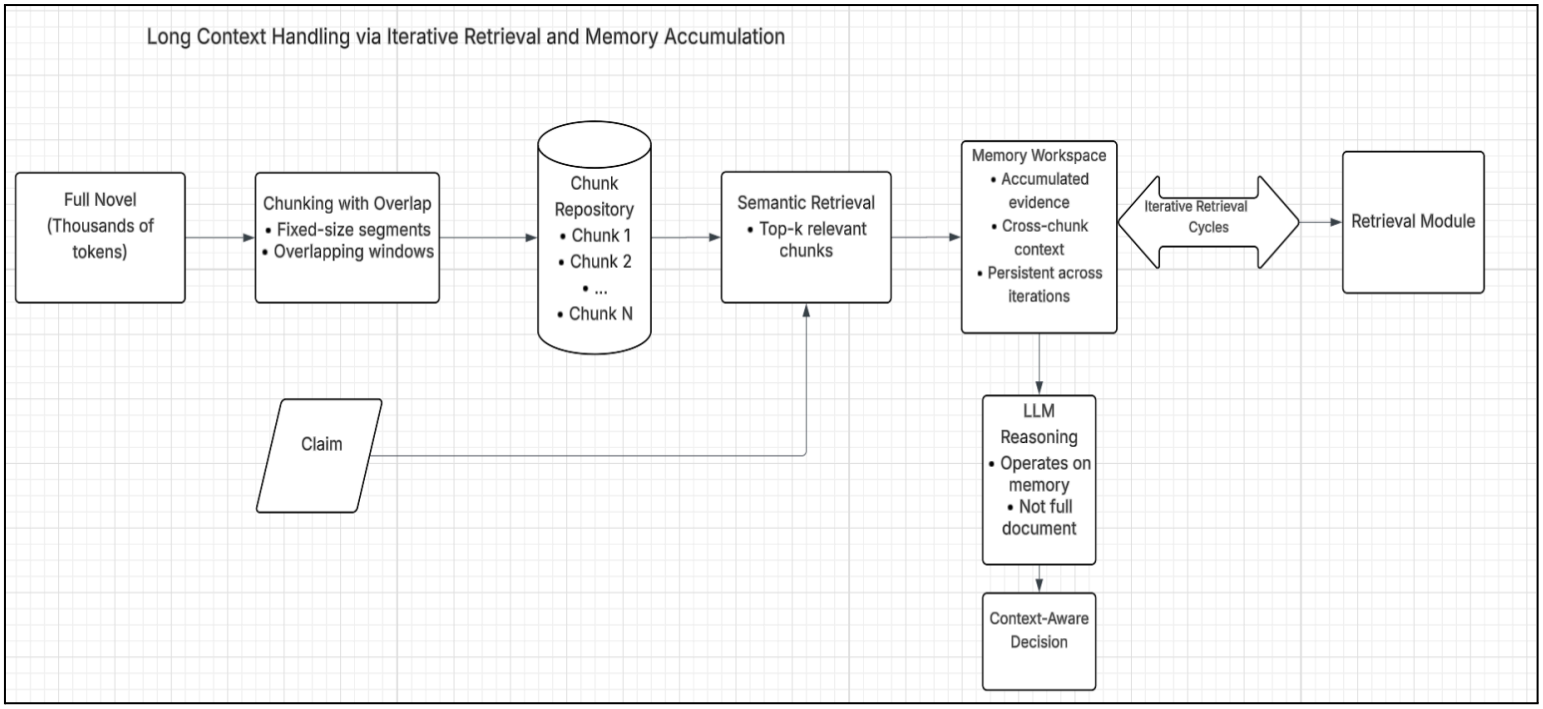


Figure 2: Proposed Architecture Diagram to handle Long Context

Figure 2 illustrates how the system processes novel-length documents using iterative retrieval and memory accumulation rather than single-pass context injection.

a. Chunking with Overlap

- The novel is split into fixed-size overlapping chunks.
- Overlap preserves narrative continuity across boundaries.

This enables scalable processing while retaining causal and temporal links that span adjacent segments.

b. Semantic Retrieval over Chunk Repository

- Each chunk is embedded and indexed in a vector store.
- Claims retrieve top-k semantically relevant chunks.

Semantic retrieval allows indirect and paraphrased evidence to be located, which is essential for narrative texts with low lexical overlap.

c. Persistent Memory Workspace

- Retrieved chunks are stored in a memory workspace.
- Evidence is accumulated across retrieval cycles.

Unlike standard RAG, evidence is not discarded after one step, enabling cross-chapter and global reasoning.

d. Iterative Retrieval and Refinement

- If evidence is insufficient, probing queries are generated.
- Retrieval is repeated to expand memory coverage.

This iterative process mirrors human rereading and supports multi-hop narrative reasoning.

e. Memory-Based LLM Reasoning

- The LLM reasons only over the memory workspace.
- The full novel is never passed into the prompt.

This avoids context window limits while enabling global, context-aware consistency decisions.

2.3 Distinguishing Noise from Casual Signals

Narrative texts contain a large amount of descriptive, incidental, and stylistic content that does not contribute to causal understanding. The core challenge is to isolate causally relevant evidence while suppressing narrative noise.

Our system addresses this through progressive filtering, memory reinforcement, and causal reasoning, rather than relying on raw similarity scores.

a. First-Level Noise Reduction via Semantic Scoring

- Retrieved chunks often include:
  - background descriptions
  - atmospheric detail
  - tangential events
- Each retrieved chunk is semantically scored and re-ranked.
- Low-relevance chunks are deprioritized before entering memory.

This ensures that the memory workspace is not polluted by stylistic or filler content.

b. Memory Reinforcement Across Retrieval Cycles

- Evidence that repeatedly appears across retrieval cycles is reinforced.
- Isolated or one-off chunks decay in importance.

Causal signals tend to recur across chapters, while noise is usually localized. Iterative memory accumulation naturally amplifies meaningful evidence.

c. Cross-Chunk Contextual Validation

- Individual chunks are evaluated in relation to previously stored memory.
- Chunks inconsistent with established context are discounted.

This prevents misleading local descriptions from overriding global narrative patterns.

d. Causal Sufficiency Check and Probing

- The system evaluates whether accumulated evidence is causally sufficient.
- If gaps exist:
    - targeted probing queries are generated
    - retrieval focuses on missing causal links

This avoids premature decisions based on incomplete or noisy evidence.

e. LLM-Based Narrative Reasoning

- The LLM reasons over the refined memory workspace, not raw retrievals.
- Decisions are based on:
    - consistency of motivations
    - causal coherence
    - alignment with character development

This shifts the task from surface-level matching to narrative-level reasoning.

2.4 Key Limitations and Failure Cases

Despite outperforming static RAG pipelines, the proposed iterative memory-driven framework still exhibits limitations when applied to complex narrative reasoning tasks. These limitations arise from retrieval constraints, implicit causality, and narrative ambiguity.

a. Embedding-Based Retrieval Limitations

Semantic embeddings do not always reflect narrative importance.

- Causally important events may be expressed subtly or indirectly
- Such chunks may receive low similarity scores
- Stylistically similar but narratively irrelevant chunks may be retrieved instead

As a result, critical causal evidence may enter memory late or not at all.

b. Chunking Granularity and Evidence Fragmentation

Narrative causality often spans multiple distant segments.

- A single chunk rarely contains complete causal justification
- Evidence may be split across non-adjacent chunks
- Partial retrieval leads to under-informed reasoning

This fragmentation can cause false negatives for globally consistent backstories.

c. Implicit and Thematic Causality

Many contradictions are never explicitly stated.

- Character motivations are inferred from behavior over time
- Ideological or moral shifts may be implied, not declared
- Thematic incompatibilities are difficult to encode explicitly

The system may misclassify subtly inconsistent backstories as consistent.

d. Temporal Reasoning Constraints

Some inconsistencies are temporal rather than factual.

- Events may be referenced non-linearly
- Causal preconditions may be implied across chapters
- Temporal ordering may require deep inference

Incorrect reconstruction of timelines can lead to faulty consistency judgments.

e. Iteration and Budget Constraints

The system operates under bounded retrieval cycles.

- Long-range dependencies may require more probing than allowed
- Early convergence can lock onto incomplete evidence
- Increasing iterations improves recall but raises compute cost

This trade-off limits exhaustive global reasoning at novel scale.

Common Failure Scenarios Observed

- **Cross-Chapter Dependency Failures**
  Cause and effect occur far apart in the narrative.

- **Multi-Chunk Evidence Requirements**
  Necessary evidence spans beyond retrieved memory.

- **Descriptive Noise Dominance**
  Rich narrative detail overshadows sparse causal cues.

3 Tech Stack

- **Python 3.10+** — Core language for system implementation, orchestration, and experiments
- **OpenAI API** — Primary reasoning and generation LLM
- **Pathways Vector Database** — Storage and retrieval of high-dimensional embeddings for efficient semantic search and memory access
- **Agent-Oriented Architecture** — Dedicated agents for probing, extraction, fusion, and validation
- **Gaussian Mixture Models (GMM)** — Clustering of embeddings into semantic and episodic memory groups
- **Stateful Memory Pool** — Persistent memory that accumulates knowledge across reasoning iterations
- **Triplet Extraction** — Conversion of text into (Subject, Relation, Object) representations
- **Knowledge Graph Construction** — Entity–relation graph built from extracted triplets
- **Hippograph (Hierarchical Knowledge Graph)** — Multi-level abstraction over entities and concepts
- **Semantic Summarization** — Graph- and text-aware abstraction of narrative information
- **Neural Embedding Models** — Dense vector representations for semantic similarity

4 Conclusion

Verifying the consistency of hypothetical character backstories against long narrative documents presents a challenge that conventional Retrieval-Augmented Generation (RAG) systems are not designed to handle. Unlike factual texts, novels encode meaning through distributed events, implicit motivations, temporal dependencies, and thematic development spanning hundreds of pages. Standard RAG pipelines assume localized evidence, explicit contradictions, and lexical relevance, leading to unreliable judgments when applied to narrative fiction.

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